The MEW nozzle in two dimensions

First results for the set-up of Ju, Omar, Mirihanage and Kilic

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Summary. The set-up of *Suppressing oozing during filament-based melt electrowriting* (SSRN 7582986) was simulated as a planar slice through the nozzle: PCL at 120 °C fed through a 0.4 mm nozzle at 6.5 kV, 5.5 mm above a collector moving at 2100 mm min^{-1}. Four results:

- At the feed printed in Table 1, 0.255 mm^3 s^{-1}, the melt is pushed out as a column whether the field is on or off.
- At 0.255 mm^3 min^{-1} the field shapes a meniscus that grows at exactly the feed rate; no jet can form on this grid.
- Only the per-minute feed matches the measured fibres: at 35 mm s^{-1} it lays a 12 µm fibre.
- The numbers are not grid-converged: with no wetting law, the melt leaves the nozzle through a neck set by the cell size. The field changes nothing before the melt reaches the collector, on any grid; a difference after contact seen on 25 µm cells is gone on 12.5 µm cells.

1 Model and set-up

Melt	PCL held at 120 °C. Viscosity 500 Pa s and surface tension 0.03 N m^{-1} are nominal values, not measurements; density 1080 kg m^{-3}.
Nozzle	0.4 mm bore, as in the paper; an assumed 1.0 mm tip face, 5.5 mm above the collector.
Field	6.5 kV, and 12 kV in one run. The melt conducts within 2 mm of the tip and is an insulator below that, so it cannot short the gap.
Collector	35 mm s^{-1} (2100 mm min^{-1}).
Feed	0.255 mm^3 s^{-1} as printed in Table 1, so the melt leaves the bore at 2.0 mm s^{-1}; and 0.255 mm^3 min^{-1}.
Grid	25 µm cells over an 8×6.8 mm slice, 14 µs steps. A free-surface lattice Boltzmann model of the melt (D2Q9, central moments; the air is not simulated), and Laplace's equation for the potential in the air (D2Q5). The melt surface feels the electric pull $\varepsilon_0 E^2/2$.

Matched to the experiment: the capillary numbers of the feed (34 as printed, 0.56 per minute) and of the collector (583), the Bond number (0.014), and the ratio of electric to capillary pressure, $\varepsilon_0 V^2/\gamma R = 62$ at 6.5 kV. Not matched: inertia, which is negligible in the experiment and kept small in the model (Reynolds number at most 0.32).

2 Results

2.1 The printed feed: a column, field or no field

The capillary number in the bore is 34, so viscous extrusion beats surface tension by a wide margin. On 25 µm cells, with the field on and off, the front falls at 1.13 mm s^{-1}, the melt within 1 mm of the tip settles at 0.65 mm^2 in both, and both columns reach the collector at 4.7 s, within 0.01 s of each other (Fig. 1; Section 2.4 for how these numbers move with the cell size). In the experiment this would look like oozing whatever the voltage.

2.2 The per-minute feed: a meniscus that grows at the feed rate

At 6.5 kV and at 12 kV alike the melt under the tip grows at 0.013 54 mm^2 s^{-1}, against a feed of 0.013 53 mm^2 s^{-1} in this planar model: nothing carries melt away (Fig. 2). A planar sheet carrying this feed at 35 mm s^{-1} would be 0.4 µm thick, against 25 µm cells, so no jet can form on this grid. After 8 s the meniscus reaches 0.32 mm below the tip face at 6.5 kV and 0.35 mm at 12 kV, and the strongest pull on it is 0.68 and 2.51 times the bore's capillary pressure, $\gamma/R = 150$ Pa. A flat tip concentrates the field less than a round one; the 12 kV run estimates the pull a round 1 mm tip feels at 6.5 kV.

2.3 Which feed?

The paper's fibres are 9.5 µm to 13.5 µm with the fan cooling and about 19 µm to 22 µm after nozzle cleaning. A straight fibre laid at 35 mm s^{-1} carries its cross-section times that speed, so 9.5 µm to 22 µm fibres carry 0.0025 mm^3 s^{-1} to 0.013 mm^3 s^{-1}. 0.255 mm^3 min^{-1} is 0.0043 mm^3 s^{-1}, a 12 µm fibre; 0.255 mm^3 s^{-1} would need a

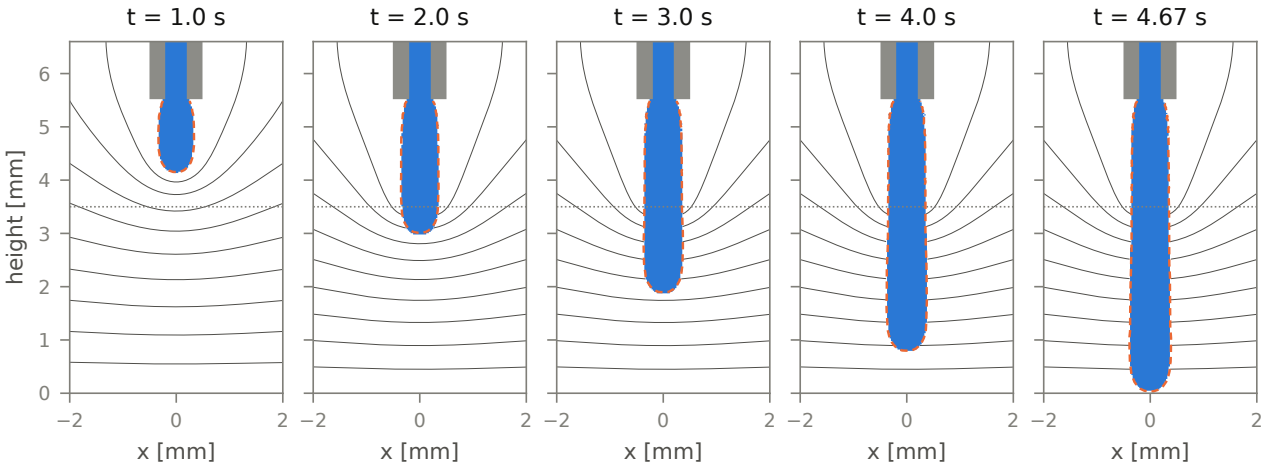


Figure 1. Table 1's feed, 0.255 $mm^3\ s^{-1}$, from the start to first contact. Blue: the melt with the field on, equipotentials every 10 % of the voltage, the melt conducting above the dotted line. Orange, dashed: the outline of the melt with the field off at the same instant. The two are almost identical.

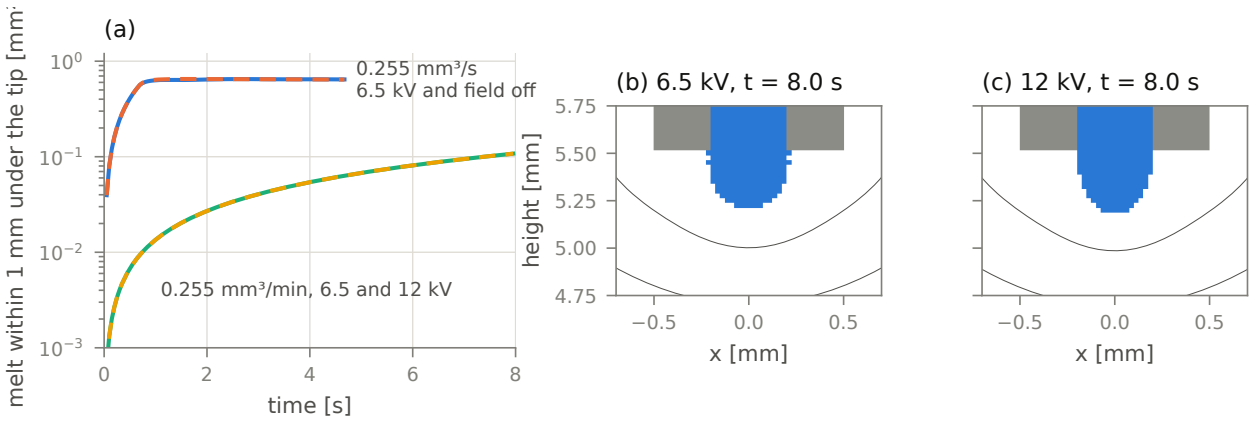


Figure 2. (a) Melt within 1 mm under the tip, the model's measure of the accumulation the paper reports as oozing. Printed feed with the field on (blue) and off (orange, dashed), to first contact; per-minute feed at 6.5 kV (aqua) and 12 kV (yellow, dashed). The pairs overlap. (b, c) The per-minute meniscus after 8 s, with equipotentials every 10 % of the voltage.

96 µm fibre.

2.4 How the numbers depend on the cell size

The printed feed was rerun on cells of 50 and 12.5 µm, with the same physics. Nothing converges: on 50, 25 and 12.5 µm cells the front falls at 0.91, 1.14 and 1.34 mm s^{-1} over the first 3.5 s, the melt within 1 mm of the tip is 0.81, 0.65 and 0.55 mm^2, and first contact comes at 5.8, 4.7 and 4.0 s. The reason is where the melt leaves the nozzle. The model has no wetting: it treats the nozzle as a surface the melt does not wet, so the melt leaves from the edge of the bore and creeps exactly two cells onto the tip face, whatever a cell measures. The neck is therefore 0.60, 0.50 and 0.45 mm wide, and since the column below carries the whole feed, a narrower neck makes a narrower, faster column. As the cells shrink the neck tends to the bore itself, 0.4 mm; a melt that wets the nozzle would hang from its outer edge instead, 1.0 mm wide, a bigger difference than any of these grids make. Before contact the field changes nothing on any grid (within 1.4 %).

2.5 After the melt reaches the collector

Both ropes leave the collector within 0.1 s of touching it. On 25 µm cells the two then part company: with the field off the rope sags back, touches again 1.2 s later near the edge of the box and is laid there, while with the field on it curls into a hook and stays up. On 12.5 µm cells that difference is gone. Both ropes rise, to 2.3 and 2.4 mm, and sink slowly, still 1.3 to 1.5 mm up 2.6 s after contact (Fig. 3); on 50 µm cells both rise too. So the field-off rope's return belongs to the 25 µm grid, not to the field, and nothing after contact here should be read as an effect of the field. Where a rope touches down is set by the width of the box in any case.

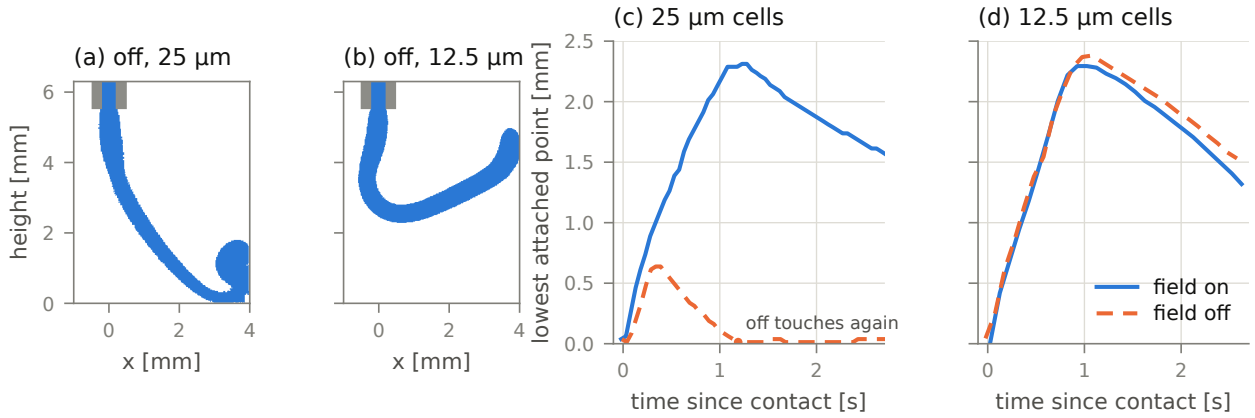


Figure 3. After first contact at the printed feed; the collector moves to the right. (a, b) The field-off rope 1.3 s after its first contact: on 25 µm cells it has come back and is laid near the edge of the box; on 12.5 µm cells it is still up. (c, d) Height of the lowest point of the melt still attached to the nozzle against the time since contact, field on (blue) and off (orange, dashed): the two part company on 25 µm cells and coincide on 12.5 µm cells.

3 What the model leaves out

- **The Taylor cone and the jet.** A planar meniscus has no conical equilibrium, and the jet is far below the grid. A flat tip also concentrates the field less than a round one.
- **Wetting.** The runs treat the nozzle and the collector as surfaces the melt does not wet. A contact-angle law since added to the code spreads the melt onto the tip face by only one cell a side in 4 s, and its contact line sticks, more so on finer grids.
- **Charge and conduction.** The melt is a perfect conductor near the tip and an insulator below; there is no free charge and no current.
- **Heat.** The melt is held at 120 °C; the hot end, where the fan acts, was modelled separately.
- **Rheology.** Newtonian at a nominal 500 Pa s: no shear-thinning, no elasticity, no die swell.

4 How far to trust it

Electric pull on a melt surface, against exact solutions	The pattern of the pull is right to 0.05 %, a moving surface's response to 3 % once its own stiffness (next row) is allowed for.
A gentle pull on a flat melt surface	Deflections of one or two cells come out 9 % to 25 % short: the free surface is too stiff there, wherever the field moves it by only a cell or two.
Surface tension: Laplace's law for a still drop	Right to 1.2 % for drops of 12 cells radius or more, 3.4 % at 8 cells. The bore is 8 cells from wall to axis.
Three cell sizes, 50, 25 and 12.5 µm	Not converged: front 0.91, 1.14 and 1.34 mm s^{-1}, melt under the tip 0.81, 0.65 and 0.55 mm^2, contact 5.8, 4.7 and 4.0 s, all set by the neck (Section 2.4); the field-off return is the 25 µm grid's.
Mass	Conserved to 0.6 % until contact; it gains up to 5.4 % while the field-off thread is drawn thin, which is not yet explained.
Symmetry	The melt is held mirror-symmetric, as the set-up is, until it touches the collector; otherwise round-off pushes the column up to 0.4 mm sideways.

5 Questions for the experiment

1. Is Table 1's feed 0.255 mm^3 s^{-1} or 0.255 mm^3 min^{-1}? At 35 mm s^{-1} the per-minute value lays a 12 µm fibre, inside the 9.5 µm to 13.5 µm measured with the fan; the per-second value would need a 96 µm fibre.
2. Are there measured values for this PCL at 120 °C: the viscosity, ideally a flow curve, and the surface tension? The model uses 500 Pa s and 0.03 N m^{-1}.
3. Does the melt ooze with the voltage off? At the printed feed the model pushes out the same column with or without the field.
4. Does the melt wet the nozzle tip: does it hang from the bore or from the nozzle's outer edge? The model assumes it does not wet, and that choice moves the column more than any grid here.

M3LB (github.com/alexderosis/M3LB, branch `electrowriting`), double precision: `demonstrator/mew_nozzle -feed s or min, -V 12, -nofield`; cells `-R 4/8/16` with `-Ucol 0.04/0.02/0.01`. Figures: `paper/mew_nozzle/make_figs.py`.